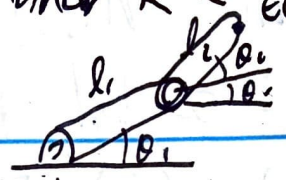
1/

2-link planar R-R EOF

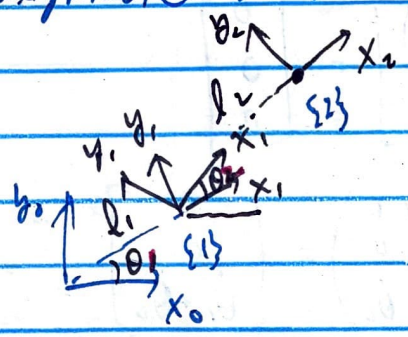
2026/2/2 (-)

YA!



Standard (SPONG)

Frame assignment {1} {2}



	a	α	d	θ
$i=1$	l_1	0	0	θ_1
$i=2$	l_2	0	0	θ_2

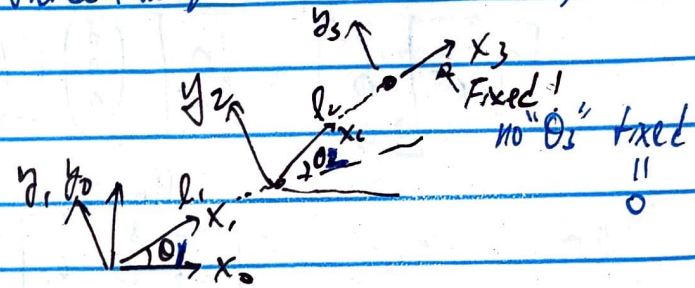
$$ {}^0_1T = \left[\begin{array}{c|c} {}^0_1R & \begin{matrix} l_1c_1 \\ l_1s_1 \\ 0 \end{matrix} \\ \hline 0\ 0\ 0 & 1 \end{array}\right] $$

Yes!

$$ {}^1_2T = \left[\begin{array}{c|c} {}^1_2R & \begin{matrix} l_2c_2 \\ l_2s_2 \\ 0 \end{matrix} \\ \hline 0\ 0\ 0 & 1 \end{array}\right] $$

Modified (Craig)

Frame Assignment {1} {2} {3}



	a	α	d	θ
$i=1$	0	0	0	θ_1
$i=2$	l_1	0	0	θ_2
$i=3$	l_2	0	0	0

$$ {}^0_1T = \left[\begin{array}{c|c} {}^0_1R & \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \\ \hline 0\ 0\ 0 & 1 \end{array}\right] $$

Yes!

$$ {}^1_2T = \left[\begin{array}{c|c} {}^1_2R & \begin{matrix} l_1 \\ 0 \\ 0 \end{matrix} \\ \hline 0\ 0\ 0 & 1 \end{array}\right] $$

moved! next operation!

$$ {}^2_3T = \left[\begin{array}{c|c} I_3 & \begin{matrix} l_2 \\ 0 \\ 0 \end{matrix} \\ \hline 0\ 0\ 0 & 1 \end{array}\right] $$

moved! next operation!

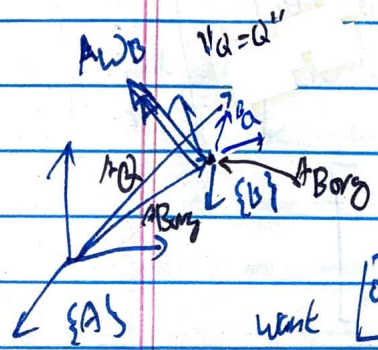
2/

Velocity & Ang. Velocity Propagation

For Both SDH & MDH.

Transpose of ${}^{i}_{i+1}R \leftrightarrow {}^{A}_{B}R$ ← origin, ← projection!

$$\begin{cases} {}^{i+1}\omega_{i+1} = \boxed{{}^{i+1}_{i}R}\,{}^{i}\omega_i + \underbrace{\dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1}}_{\text{revolute}} \\ {}^{i+1}v_{i+1} = {}^{i+1}_{i}R\left[\underbrace{{}^{i}v_i + {}^{i}\omega_i \times {}^{i}P_{i+1}}\right] + \underbrace{\dot{d}_{i+1}\,{}^{i+1}\hat{Z}_{i+1}}_{\text{prismatic}} \end{cases}$$



From ${}^{A}V_Q = {}^{A}V_{Borg} + {}^{A}_{B}R\,{}^{B}V_Q + {}^{A}\omega_B \times {}^{A}Q$, ${}^{A}_{B}R\,{}^{B}Q$

in terms of $\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\end{pmatrix}$ control!

For our 2-link R-R Planar:

want $\boxed{{}^{0}v_2}$ only @ EOF ⇒ what we want $\boxed{{}^{0}v_3}$

SDH. ⇒ {1}, {2}

$$\boxed{\begin{matrix}{}^{0}\omega_0\\ {}^{0}v_0\end{matrix}} \rightarrow \boxed{\begin{matrix}{}^{1}\omega_1\\ {}^{1}v_1\end{matrix}} \rightarrow \boxed{\begin{matrix}{}^{2}\omega_2\\ {}^{2}v_2\end{matrix}}$$

MDH. ⇒ {1}, {2}, {3}

$$\boxed{\begin{matrix}{}^{0}\omega_0\\ {}^{0}v_0\end{matrix}} \rightarrow \boxed{\begin{matrix}{}^{1}\omega_1\\ {}^{1}v_1\end{matrix}} \rightarrow \boxed{\begin{matrix}{}^{2}\omega_2\\ {}^{2}v_2\end{matrix}} \rightarrow \boxed{\begin{matrix}{}^{3}\omega_3\\ {}^{3}v_3\end{matrix}}$$

Find Jacobian! Geometrically!

recall ${}^{0}_{1}T = \left[\begin{array}{c|c} {}^{0}_{1}R & \begin{matrix} l_1 c1 \\ l_1 s1 \\ 0 \end{matrix} \\ \hline 0\,0\,0 & 1 \end{array}\right]$

${}^{1}_{2}T = \left[\begin{array}{c|c} {}^{1}_{2}R & \begin{matrix} l_2 c2 \\ l_2 s2 \\ 0 \end{matrix} \\ \hline 0\,0\,0 & 1 \end{array}\right]$

$$\boxed{\text{Same } {}^{1}_{0}R = \begin{bmatrix} c1 & s1 & 0 \\ -s1 & c1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

${}^{0}_{1}T = \left[\begin{array}{c|c} {}^{0}_{1}R & \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \\ \hline 0\,0\,0 & 1 \end{array}\right]$

${}^{1}_{2}T = \left[\begin{array}{c|c} {}^{1}_{2}R & \begin{matrix} l_1 \\ 0 \\ 0 \end{matrix} \\ \hline 0\,0\,0 & \phi \end{array}\right]$

$$\boxed{\text{Same } {}^{2}_{1}R = \begin{bmatrix} c2 & s2 & 0 \\ -s2 & c2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

${}^{2}_{3}T = \left[\begin{array}{c|c} I_3 & \begin{matrix} l_2 \\ 0 \\ 0 \end{matrix} \\ \hline 0\,0\,0 & 0 \end{array}\right]$

☆ common Rotational matrix: SDH {1} R+T, MDH {1} R+0, same thing @ {2}

$${}^{0}_{1}R = \begin{bmatrix} c1 & -s1 & 0 \\ s1 & c1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^{1}_{2}R = \begin{bmatrix} c2 & -s2 & 0 \\ s2 & c2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3/ derivation of MDH

2-link planar R-R

Step 1

$$^1\omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 \end{bmatrix} = {}^1_0R\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 \end{bmatrix} \qquad {}^0\omega_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad {}^0v_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$^1v_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = {}^1_0R\left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}\right] = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2.

$$^2\omega_2 = \underbrace{\begin{bmatrix} c2 & s2 & 0 \\ -s2 & c2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{({}^2_1R)}\begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{bmatrix}$$

$$^2v_2 = \underbrace{\begin{bmatrix} c2 & s2 & 0 \\ -s2 & c2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{({}^2_1R)}\left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 \end{pmatrix} \times \begin{pmatrix} l_1 \\ 0 \\ 0 \end{pmatrix}\right] = \begin{bmatrix} c2 & s2 & 0 \\ -s2 & c2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ l_1\dot\theta_1 \\ 0 \end{bmatrix} = \begin{bmatrix} l_1 s2\dot\theta_1 \\ l_1 c2\dot\theta_1 \\ 0 \end{bmatrix}$$

Step 3.

$$^3\omega_3 = [I_3]\begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{bmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_3 = 0 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{bmatrix}$$

$$^3v_3 = [I_3]\left[\begin{pmatrix} l_1 s_2\dot\theta_1 \\ l_1 c_2\dot\theta_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{pmatrix} \times \begin{pmatrix} l_2 \\ 0 \\ 0 \end{pmatrix}\right] = \left[\begin{pmatrix} l_1 s_2\dot\theta_1 \\ l_1 c_1\dot\theta_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ l_2(\dot\theta_1 + \dot\theta_2) \end{pmatrix}\right]$$

$$^0v_3 = \underbrace{{}^0_1T\,{}^1_2T\,{}^2_3T}_{\text{not necessary}}\,{}^3v_3, \quad \text{or} \quad {}^0_3R = {}^0_1R\,{}^1_2R\,{}^2_3R = \begin{bmatrix} c12 & -s12 & 0 \\ s12 & c12 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3\text{ by }3)$$

$$\boxed{\begin{aligned} &sACB - cASB \\ &= s(A-B) \end{aligned}}$$

$$^0v_3 = \begin{bmatrix} c12 & -s12 & 0 \\ s12 & c12 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} l_1 s2\dot\theta_1 \\ l_1 c2\dot\theta_1 + l_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix} = \begin{bmatrix} l_1 s2 c12\dot\theta_1 - l_1 c2 s12\dot\theta_1 - l_2 s12(\dot\theta_1 + \dot\theta_2) \\ l_1 s2 s12\dot\theta_1 + l_1 c2 c12\dot\theta_1 + l_2 c12(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix}$$

$$\boxed{\begin{aligned} &sASB + cACB \\ &= c(A-B) \end{aligned}}$$

$\sin\theta_2\cos(\theta_1+\theta_2) - \cos\theta_2\sin(\theta_1+\theta_2)$
$= \sin(\theta_2 - \theta_1 - \theta_2) = -\sin\theta_1$

$\sin\theta_2\sin(\theta_1+\theta_2) + \cos\theta_2\cos(\theta_1+\theta_2)$
$= \cos(\theta_2 - \theta_1 - \theta_2) = \cos\theta_1$

$$= \begin{bmatrix} -l_1\sin\theta_1\cdot\dot\theta_1 - l_2\sin(\theta_1+\theta_2)(\dot\theta_1 + \dot\theta_2) \\ l_1\cos\theta_1\cdot\dot\theta_1 + l_2\cos(\theta_1+\theta_2)(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix}$$

4/ derivation on SDH $\{1\}, \{2\}$ only

2-link Planar R-R.

$$^0_1T = \left(\begin{array}{c|c} {}^0_1R & \begin{matrix} l_1C_1 \\ l_1S_1 \end{matrix} \\ \hline -0- & 1 \end{array}\right) \qquad {}^1\omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 \end{bmatrix} \qquad {}^0\omega_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \; {}^0v_0 = 0$$

$$^1_2T = \left(\begin{array}{c|c} {}^1_2R & \begin{matrix} l_2C_2 \\ l_2S_2 \\ 0 \end{matrix} \\ \hline -0- & 1 \end{array}\right)$$

$$^1v_1 = \begin{bmatrix} C_1 & S_1 & 0 \\ -S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 \end{pmatrix} \times \begin{pmatrix} l_1C_1 \\ l_1S_1 \\ 0 \end{pmatrix} \right] = \begin{bmatrix} C_1 & S_1 & 0 \\ -S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -l_1S_1\dot\theta_1 \\ l_1C_1\dot\theta_1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -l_1S_1C_1\dot\theta_1 + l_1S_1C_1\dot\theta_1 \\ l_1S_1^2\dot\theta_1 + l_1C_1^2\dot\theta_1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ l_1\dot\theta_1 \\ 0 \end{bmatrix}$$

$$^2\omega_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot\theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{bmatrix}$$

$$^2v_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{pmatrix} 0 \\ l_1\dot\theta_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot\theta_1 + \dot\theta_2 \end{pmatrix} \times \begin{pmatrix} l_2C_2 \\ l_2S_2 \\ 0 \end{pmatrix} \right]$$

$$= \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{pmatrix} 0 \\ l_1\dot\theta_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -l_2S_2(\dot\theta_1 + \dot\theta_2) \\ l_2C_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{pmatrix} \right] = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -l_2S_2(\dot\theta_1 + \dot\theta_2) \\ l_1\dot\theta_1 + l_2C_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cancel{-l_2S_2C_2(\dot\theta_1 + \dot\theta_2)} + l_1S_2\dot\theta_1 + \cancel{l_2S_2C_2(\dot\theta_1 + \dot\theta_2)} \\ \underbrace{+ l_2S_2S_2(\dot\theta_1 + \dot\theta_2)}_{S_2^2} + \overset{+}{l_1C_2\dot\theta_1} + \underbrace{l_2C_2C_2(\dot\theta_1 + \dot\theta_2)}_{C_2^2 = 1} \\ 0 \end{bmatrix} = \begin{bmatrix} l_1S_2\dot\theta_1 \\ l_1C_2\dot\theta_1 + l_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix}$$

Since $^0_2R = {}^0_1R\,{}^1_2R = \text{same} = \begin{bmatrix} C_2 & -S_2 & 0 \\ S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\underset{SDH}{^2v_2} = \underset{MDH}{^3v_3}$$

$$\Rightarrow {}^0v_2 = {}^0_2R\,{}^2v_2 = \begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} l_1S_2\dot\theta_1 \\ l_1C_2\dot\theta_1 + l_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix} = \begin{bmatrix} -l_1S_1\dot\theta_1 - l_2S_2(\dot\theta_1 + \dot\theta_2) \\ l_1C_1\dot\theta_1 + l_2C_2(\dot\theta_1 + \dot\theta_2) \\ 0 \end{bmatrix}$$